Chi squared goodness of fit tests

1 The steps for testing a hypothesis with a χ^2 GOF test are listed below in the wrong order. Put the list into the correct order.

- **A:** Calculate the value of the test statistic χ^2 and p-value using the calculator.
- **B:** If $p \leq \alpha$ then state that H_0 is rejected.
- **C:** State the significance level, α.
- **D:** State the null hypothesis and alternative hypothesis.
- **E:** Put the observed frequencies in List 1 and the expected frequencies in List 2 of the Statistics application on the calculator.
- **F:** Calculate the degrees of freedom, (df), for the data.
- **G:** Calculate the expected frequencies for the data.

2 A school conducted a survey of its year 12 students to collect health information. The survey revealed that a substantial proportion of students were not engaging in regular exercise and many felt their nutrition was poor. In response to a question on regular exercise, 55% of all students reported getting no regular exercise, 30% reported exercising occasionally and 15% reported exercising regularly. The next year the school launched a health promotion campaign on campus in an attempt to increase health behaviours among year 12's. The program included posters and flyers on exercise and nutrition. To evaluate the impact of the campaign, the school again surveyed students and asked the same questions. The survey was completed by 210 students and the following data was collected regarding the regularity of exercise:

	No regular exercise	Occasional exercise	Regular exercise	Total
Observed number of students	113	72	25	210
Expected number of students				

In order to assess whether the health promotion campaign has been successful in improving exercise levels among year 12's the school conducts a χ^2 goodness of fit test at a 5% significance level.

a State the hypotheses set for this problem.

b State the level of significance, α, as a decimal number.

c Complete the table above with the expected numbers of students.

d State the number of degrees of freedom (df) for this data.

e Use your calculator to find the χ^2 and p-value, correct to four decimal places.

f Comment on your findings.

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3 The colours in a packet of smarties are said to be evenly distributed by the manufacturer. James conducts a large scale experiment and from 5428 smarties he finds the distribution of colours as follows:

	Pink	Blue	Red	Yellow	Orange	Brown	Purple	Green
Observed freq	678	739	797	714	566	556	725	653

James decides to test if this sample is consistent with the manufacturer's claim using a χ^2 goodness of fit test at a 5% significance level.

a Define the hypothesis set for the test.

b State the level of significance, α, as a decimal number.

c State the expected frequency for each colour of smartie.

d State the number of degrees of freedom (df) for this data.

e Use your calculator to find the value of χ^2 correct to two decimal places.

f Use your calculator to find the value of p, expressing your answer in scientific notation.

g Comment on your findings.

4 Scott is testing a die with eight sides to see if it is fair. The results of 88 rolls of the die are given in the table below:

	1	2	3	4	5	6	7	8
Observed freq	8	11	12	10	13	11	15	8

Scott decides use a χ^2 goodness of fit test at a 2% significance level.

a Define the hypothesis set for the test.

b State the level of significance, α, as a decimal number.

c Complete the table of observed and expected frequencies:

	1	2	3	4	5	6	7	8
Observed freq	8	11	12	10	13	11	15	8
Expected freq								

d State the number of degrees of freedom (df) for this data.

e Use your calculator to find the value of χ^2 correct to two decimal places.

f Use your calculator to find the value of p, correct to two decimal places.

g Comment on your findings.

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5 The proportions of blood types, O, A, B and AB are known to be in the ratio $47 : 37 : 10 : 6$ in a particular city. Researchers take samples from students at a school within the city and find the distribution of blood types to be as follows:

	O	A	B	AB
Observed frequency	88	57	21	6

They decide to use a χ^2 goodness of fit test at a 10% significance level to determine whether the blood type proportions at the school differ significantly from the proportions in the wider city.

a Define the hypothesis set for the test.

b State the level of significance, α, as a decimal number.

c Complete the table of observed and expected frequencies:

	O	A	B	AB
Observed frequency	88	57	21	6
Expected frequency				

d State the number of degrees of freedom (df) for this data.

e Use your calculator to find the value of χ^2 correct to two decimal places.

f Use your calculator to find the value of p, correct to two decimal places.

g Comment on your findings.

6 A circular spinner is divided into three unequal parts. The green sector takes up an angle of 250° at the centre. The red sector takes up an angle of 60° at the centre and the blue sector takes up the remainder of the spinner. The spinner is spun 200 times and the results are as follows:

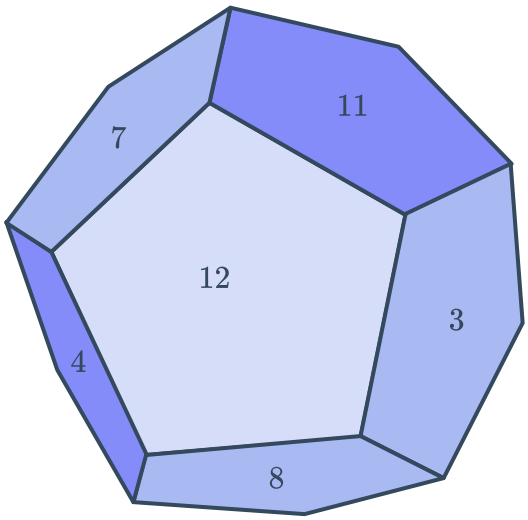
	Green	Red	Blue
Observed frequency	142	31	27

Perform a χ^2 goodness of fit test at a 5% significance level to determine whether the spinner appears to be fair.
When conducting your test be sure to state the following:

- Hypotheses and significance level.
- Observed and expected frequencies.
- df, χ^2 and p-value.
- A comment on whether the hypothesis is rejected or not rejected.

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7 A 12-sided die has faces with the numbers 1 through 12 as shown:
The die is rolled 300 times and gives the following results:



	< 4	$4 - 9$	> 9
Observed freq	59	180	61

Perform a χ^2 goodness of fit test at a 5% significance level to determine whether the die appears to be fair.

When conducting your test be sure to state the following:

- Hypotheses and significance level.
- Observed and expected frequencies.
- df, χ^2 and p-value.
- A comment on whether the hypothesis is rejected or not rejected.

8 An insurance company proposes that the chance of a car accident happening on the weekend is double that of an accident happening during the week. Data on car accident frequency is collected and displayed in the table below:

	Mon	Tue	Wed	Thurs	Fri	Sat	Sun
Observed freq	25	19	28	22	18	39	45

Conduct a χ^2 goodness of fit test at a 2% significance level to test the hypothesis.
When conducting your test be sure to state the following:

- Hypotheses and significance level.
- Observed and expected frequencies.
- df, χ^2 and p-value.
- A comment on whether the hypothesis is rejected or not rejected.

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Critical values

9 For each of the following determine whether the null hypothesis should be rejected or not:

a

- $H_0 : p_1 = \frac{1}{3}, p_2 = \frac{1}{3}, p_3 = \frac{1}{3}$
- H_1 : at least one of $p_1 \neq \frac{1}{3}, p_2 \neq \frac{1}{3}, p_3 \neq \frac{1}{3}$
- χ^2 calculated value is 4.56
- χ^2 critical value is 5.99

b

- $H_0 : p_1 = \frac{1}{4}, p_2 = \frac{1}{8}, p_3 = \frac{1}{4}, p_4 = \frac{3}{8}$
- H_1 : at least one of $p_1 \neq \frac{1}{4}, p_2 \neq \frac{1}{8}, p_3 \neq \frac{1}{4}, p_4 \neq \frac{3}{8}$
- χ^2 calculated value is 11.14
- χ^2 critical value is 6.25

10 A share trading platform asked users to rate their service and found the distribution of ratings to be as follows:

	Poor	Average	Good	Excellent
Percentage	37%	25%	26%	12%

They are not happy with the ratings and initiate some changes in their online platform. One year later they survey customers again and record the following data:

	Poor	Average	Good	Excellent
Number of customers	26	67	88	110

They decide to use a χ^2 goodness of fit test at a 5% significance level to determine whether the rating proportions differ significantly after the changes.

a Define the hypothesis set for the test.

b Complete the table of observed and expected frequencies:

	Poor	Average	Good	Excellent
Observed frequency	26	67	88	110
Expected frequency				

c State the number of degrees of freedom (df) for this data.

d Use your calculator to find the value of χ^2 correct to two decimal places.

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e If the critical value of χ^2 is 7.81, comment on whether the implemented changes appear to be beneficial.

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